17 Polysemy and the Count–Mass Distinction: What Can We Derive from a Lexicon of Count and Mass Senses?

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1. Introduction

While there is some controversy in the literature on the count–mass distinction concerning its lexical base (with some proponents assuming that the count–mass distinction is not a lexical distinction at all), it is undisputed that lexical items in one way or another are affected by the count–mass distinction. From this perspective, it is remarkable that current research on the count–mass distinction does not elaborate on two aspects that might provide new insights: *lexical variation* and *lexical ambiguity*. We borrow the term *lexical variation* from experimental linguistics, where it is stated that experiments should make use of a certain number of lexical items, of which the linguist implicitly assumes that they belong to the same class. With the exception of Allan (1980), lexical variation does not figure prominently in dealing with the count–mass distinction, and hence the count–mass distinction as a binary one is mostly maintained without question. Most analyses, including influential works such as Pelletier (1975), Link (1983), Bunt (1985), Krifka (1989), Landman (1989, 2011, 2016), Pelletier and Schubert (1989/2003), Chierchia (1998a; 2010; this volume, Chapter 2), and Rothstein (2010, 2017; this volume, Chapter 3), among others, rest on a small set of so-called celebrity nouns (the term comes from an unpublished talk by Scott Grimm at the 2012 LSA meetings) such as *table*, *horse*, *cat*, *child*, *ring*, *cow*, *children*, *water*, *sand*, *mud*, *wine*, *footwear*, *gold*, *furniture* being representative for the whole class of nominals to which the count–mass distinction applies, to which recently (cf. Zucchi and White 2001; Rothstein 2010) *twig*, *rope*, and *bouquet* have been added (or rediscovered, since the properties of these nouns had already been discussed in Krifka 1989). These analyses tacitly assume that what is said about the respective celebrities carries over to the other members of the respective countability class (which includes the assumption that there are only two classes with a few notorious exceptions, such as *furniture* or *twig*). In the following, we will show that a deeper look into the data will reveal variation that remains unaccounted for, and that in addition, ambiguity with regard to countability differences raises further questions.

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Lexical ambiguity does not actually need an introduction, as every linguist knows about the pitfalls and problems of homonymy and polysemy (not the least being the question of how to disentangle the aforementioned terms). While the early discussion of the count–mass distinction within formal semantics acknowledged the role of ambiguity (cf. Pelletier 1975), lexical ambiguity was only slowly taken for granted and even considered as irrelevant, as the discussion of the count–mass distinction progressed. A famous example of a problematic treatment of ambiguity in the count–mass distinction can be found in Payne and Huddleston (2002: 334–5), where the authors show "polysemy" between count and mass senses. Some of the examples they provide, however, make use of a form that should be categorized as a noun in one sense, but as a part of a support verb construction (1a,b) or even as a participle (1c,d) in the other.

(1) a. This proposal has three *advantages*. [count]
b. They took *advantage* of us. [mass according to Payne and Huddleston]
c. It is certainly a fine *building*. [count]
d. There's plenty of *building* going on. [mass according to Payne and Huddleston]

It is highly questionable that we are dealing with a case of ambiguity here that has any bearing on the count–mass distinction (and particularly so in (1c,d)). But there are other, quite common cases of dealing with ambiguity in a somewhat agnostic way. Two typical cases are so-called *dual-life* nouns (Pelletier and Schubert 1989) and a set of countability shifts including cases of *grinding* (Pelletier 1975), *packaging* (Bach 1986), and *sorting* (Bunt 1985).

As for the former case, Pelletier (2012) assumes that the noun *chocolate* has the same meaning (modulo individuation vs. stuff) in (2a) and (2b). In the first case, *chocolate* is shown in a mass context, while in the second case, it appears in a count context:

(2) a. a lot of chocolate
b. many more chocolates

The question regarding the nouns in (2) is: what should remain as the common semantic core of *chocolate* in (2a) and (2b) that actually justifies the claim that the noun and its semantics remain constant while undergoing a shift in countability? Pelletier remarks that *chocolates* have to be in a confection to be used as in (2b), but no such constraint applies to *chocolate* in (2a). In addition, one might argue that *chocolates* do not necessarily have to contain *chocolate*, which again raises the question of the identity of the two nouns in (2).[1] The

[1] It is interesting to observe how different languages lexically realize the concepts in question. While German draws a clear distinction between *Schokolade* and *Pralinen*, French uses *chocolat* vs. *bouchée au chocolat* or even *offrir de chocolats* (with a plural, meaning: *to give away chocolates*).

original grinder explanation in Pelletier (1975) was restricted to objects designated by concrete nouns, and it generated "undifferentiated stuff" that contained all the materials that comprised the starting object. So, if one supplied an octopus to the grinder, one would end up with a gooey mess that contained the total remnants of the octopus – eyeballs, beak, suction cups, and all. Clearly this is different from (for example) the edible portions of an octopus. And (3b) and (3c) are different, only the former being an application of a grinder-like operation envisaged by Pelletier (1975) on the meaning given by the WordNet sense in (3a) (Miller 1995; Fellbaum 1998).

(3) a. armadillo#1: burrowing, chiefly nocturnal mammal with body covered with strong horny plates.
b. There is armadillo all over the road
c. We're having armadillo for dinner this evening.

Some authors use the notion of "grinding" for both of these meanings, and even more generally for any way of generating a mass meaning from a count meaning.[2] The following discussion of grinding in Rothstein (2010: 390–4) further illustrates the point. Referring to Pelletier and Schubert (1989), Rothstein starts by considering examples like (4), and proposes the rule $\text{SHIFT}_{\text{mass}}$ to turn count nouns into "ground" mass nouns:

(4) a. After he had finished the job, there was bicycle all over the floor.
b. After the accident, there was boy all over the ground.

(5) $\lambda P \lambda x.\exists y[y \in \pi_1(P) \wedge x \leq y \wedge \neg x = y]$

What (5) says is that one can take the first projection of a count predicate (i.e. the set of atoms that are contextually determined as such, since Rothstein (2010) assumes that count predicates are of type $<<e \times k>, t>$)[3] and turn this predicate into a predicate denoting a set of proper parts of the atoms. It is already unclear why the set of parts must be proper, i.e. not consisting of everything the atom was made of, but even leaving this aside, this would again only account for the contrast between (3a) and (3b).

Clearly, in the (3c) case we are employing a different sense of *armadillo* from the one used in (3a) – indeed, even the denotation of the two occurrences of *armadillo* are different. The case is not quite so clear in contrasting (3b)

[2] For example, Chierchia (2010: 106) uses "universal grinder" to massify a count noun N. Chierchia (this volume, Chapter 2) takes grinding to be the main count to mass application, and thus assumes: "For any count property P_C, $G(P_C)$, if defined, is the mass property corresponding to P_C" (p. 31).

[3] Rothstein (2010) actually uses type $<d \times k, t>$ following Landman (1996) who uses type *d* for entities and type *e* for events. We have converted this to the more usual convention: *e* for entities and *v* for events. Cf. also the discussion below in Section 5.

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with (3a), since at least all the parts of the relevant armadillo are the same, even if the structural integrity differs. Here, one needs to take into consideration that the outcome of the Universal Grinder is indeed a difference in structural integrity, whereas shifts from mass to count, as for instance (3c) and (1b), are different. As for Rothstein's approach (which remains one of the very few to actually provide a formal account of grinding), we note that it makes a wrong prediction with regard to (3a) and (3c), for even in the case that we would like to have *armadillo* for dinner tonight, we might not expect all arbitrary parts of the armadillo to be served (which, counter our intuitions, would be sufficient if Rothstein's grinder would apply to (3a,c)). By means of many more examples that we will discuss in this paper, we will clarify that it is not possible to narrow down all marked mass uses of nouns (either accounted for as specific senses or coerced meanings) to an operation of grinding, no matter how widely understood, since the core function underlying grinding, viz. transforming an object to ground stuff of all the ingredients of that object, is not what is happening with the denotation of nouns used in mass contexts, such as (3c) and (1b).

As an illustration of various mass uses of nouns that do not result from grinding, consider the following more realistic scenario. A visitor of the swamps of Florida appears to be hungry to the person he visits, so that person remarks that *there is alligator in the fridge*. Of course, the visitor will not expect to be attacked by an alligator after opening the fridge (because *alligator$_{mass}$* refers to meat, not to the animal). Nor would the visitor expect a ground up totality of alligator – the scales, claws, teeth. In sum, the aforementioned examples show a difference between count and mass, but also a difference between two related senses. Landman (2011) has argued that *grinding* should not be used as an argument for or against possible analyses of the count–mass distinction, because there will always be a change in a sense that has nothing to do with the distinction between count and mass proper. By the same line of reasoning, Quirk et al. (1985: 1564) dismiss grinding in favor of characterizing examples like *an inch of pencil* or *a few square feet of floor* as instances of a polysemy that they call "N viewed in terms of measurable extent."

This paper follows the same lead but tries to systematize different cases of polysemy in the area of the count–mass distinction. We will argue below that the majority of cases called grinding are actually realizations of a regular polysemy (cf. Falkum 2010).

Lexical ambiguity within the count–mass distinction also has an inhibiting effect on empirical research such as corpus studies. Unless a corpus is fully annotated for different meanings of the same noun form, a quantitative study may come to the conclusion that count occurrences and mass occurrences of the very same noun occur in equal numbers. As an illustration, consider the case of *alarm*, which receives the following four definitions in WordNet (Fellbaum 1998):

(6) *alarm*:
a. fear resulting from the awareness of danger
b. a clock that wakes a sleeper at some preset time
c. a device that signals the occurrence of some undesirable event
d. an automatic signal (usually a sound) warning of danger

We do not want to engage in a discussion of whether the definitions provided in (6) are the result of polysemy or homonymy, but crucially, *alarm* in (6a) is taken to be a *mass* noun, while the other three definitions refer to *count* uses. With a purely structural analysis, i.e. without further *semantic* annotations in the corpus, it is likely that the count and mass senses of *alarm* cannot be discerned, and hence wrong conclusions are drawn with respect to the count and mass occurrences of *alarm*, even up to the point where *alarm* is assigned the status of a *dual-life noun*. In order to add some structure to discussions of the term *dual-life noun*, we will speak of *proper dual life nouns* when we refer to nouns that show (at least) two discernible senses, where the difference between the senses only pertains to the count–mass distinction. It should be obvious from the definitions provided in (6) that *alarm* is not a *proper dual-life noun*:[4] while the different senses listed in (6) may fall on either side of a count–mass distinction, they also exhibit different senses irrespective of the count–mass distinction.

This paper addresses the aforementioned methodological problem by introducing the *Bochum English Countability Lexicon (BECL)* (Kiss, Pelletier, and Stadtfeld 2014; Kiss et al. 2016). BECL is a database of approximately 11,000 count and mass nouns, based on syntactic and semantic annotations of pairs of a noun form with a specific sense derived from WordNet (Fellbaum 1998). BECL addresses *lexical variation* and *lexical ambiguity* at the same time. Of course, BECL – as a lexicon – does not free the researcher from the task of annotating occurrences of count and mass nouns in a corpus, but it guides the researcher in various ways. So, BECL, which is based on the sense definitions of WordNet (for instance containing information about count and mass types related to senses), can thus be used to systematically extract contexts for individual words, which despite other ambiguities would only occur as count or mass nouns respectively. If deviations from this pattern are found, this may imply that the set of definitions was not complete or may lead to further characterization of systematic polysemy in the count–mass distinction. BECL also provides indications for further research that are not easily accessible if the empirical base is formed only by the aforementioned celebrity

[4] And perhaps should not even be considered a *dual-life noun*.

nouns. As an example, in BECL, e.g., we find about 3.5 times more count than mass noun sense types.[5]

In Section 2, we will briefly introduce BECL. Section 2 will also address the distribution of count and mass types within BECL and its implications. Section 3 presents four different kinds of ambiguity found in BECL, the last type of which is called T4 ambiguity, which, we believe, presents a particularly interesting case in light of the count–mass distinction, because according to the annotations, it looks as if these nouns show a unique semantics that differs only with regard to the count–mass distinction – hence the relevant nouns are *proper dual life nouns*. Importantly, this is not the case with common count–mass variations such as *lamb* or *rabbit*. T4 ambiguity will be dealt with in Section 4. Section 5 will address implications derived from BECL. We conclude by assuming that our findings are generally compatible with theories of the count–mass distinction that place this distinction in the semantics of nouns (such as Chierchia (1998a; 2010; this volume, Chapter 2) and Rothstein (2010; this volume, Chapter 3)), although these theories need to disentangle lexical ambiguity and not lump all cases together: dual-life, proper dual-life, and coercion in one pot. Furthermore, our findings provide clear challenges for theories that assume that lexical nouns are neutral with respect to the count–mass distinction, and that syntactic contexts mainly determine whether nouns are count or mass (such as Borer (2005) and Pelletier (1975, 2012)).

2. The Bochum English Countability Lexicon

The Bochum English Countability Lexicon (BECL) is a lexicon of English noun–sense pairs enriched with countability information. Four native speakers of Canadian English have annotated a set of pairs consisting of a noun form and a sense with regard to their behavior in four syntactic and two semantic patterns. The nouns have been derived in an arbitrary fashion from frequently occurring nouns in the Open American National Corpus (www.anc.org), for which sense definitions are provided in WordNet (Fellbaum 1998).[6]

The annotators had to determine whether (a) the noun–sense pair may occur in the syntactic contexts provided by four syntactic tests, and (b) depending on

[5] Here and in the following we define a *noun sense type* as a type of noun with a specific sense. Since BECL is a lexicon, it does not contain noun sense tokens, but only noun sense types. The observation is thus that the lexicon contains about 3.5 times more pairs of a noun and a sense that are classified as count than pairs that are classified as mass. We do not make any claim about the frequency of occurrence in natural language data but would like to point out that the noun forms chosen to derive BECL were drawn from high frequency noun forms in the Open American National Corpus.

[6] It should be noted that our pragmatic reliance on WordNet does not necessarily entail that we think that all senses come neatly cut and packaged. Indeed, we are aware of the various shortcomings that are part of the art and science of lexicography.

Table 17.1. *Major classes in BECL*

major class	example	sum
count	academy#3 (a school for special training)	8,390
mass	pride#1 (a feeling of self-respect and personal worth)	2,405
count and mass	ambition#2 (a strong drive for success)	698
neither mass nor count	commons#4 (the common people)	141
		11,634

the outcome of the first two syntactic tests, had to answer two questions about the interpretation of the resulting constructions.[7] A combination of a noun and a sense was included in the classification if it received identical annotations from independent annotators. A total number of over 18,000 noun–sense pairs have been annotated by up to four annotators.[8] From these, we derived the approximately 11,000 noun–sense pairs that form BECL (the current version being 2.1). We have collected the independent annotations and classified them so that four major and eighteen sub-classes could be identified. The four major classes are described in Table 17.1: (a) regular count, (b) regular mass, (c) count and mass, and (d) noun–sense pairs that are neither count nor mass.[9]

The general structure of BECL thus indicates that the concepts *count noun* and *mass noun* are discernible, even if their basic characteristics are unclear. If the sample of nouns we have annotated is approximately representative,[10] then it follows that the majority of English noun–sense pairs are uniquely count; another large group (which, however, is much smaller than the count noun group) is mass; and we find two further, significantly smaller groups that can be classified as either being *both* mass and count or *neither* mass *nor* count. We will refer to these four groups (and also to subgroups of these four groups) as *countability classes*. Each countability class contains further sub-classes,

[7] For more details on the annotation process we refer to Kiss, Pelletier, and Stadtfeld (2014) and Kiss et al. (2016), which include a documentation of syntactic and semantic tests used in the annotation process. The tests are also discussed briefly in Section 4.

[8] The majority of the data was annotated by two independent speakers of Canadian English. One part of the data (approx. 1000 noun senses) was given to all four annotators as a trial test of inter-annotator agreement. The noun senses that ended up in BECL were annotated unanimously by all annotators, either two or four.

[9] We named the classes according to the outputs of the annotation process. A noun–sense pair which is both mass and count means that the annotations indicate count and mass features (as will become clear in Section 4, when T4 is discussed). Importantly, a noun-sense from the final major class, neither mass nor count, is not meant to be a noun without countability, but rather a noun-sense that does not show any particular tendencies towards a classification as count or as mass.

[10] We are aware, of course, that no corpus is representative, so what we can claim here is basically that the nouns derived from the corpus have been derived in a sufficiently arbitrary way.

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which emerge from different options in the answers the annotators have chosen. In total, BECL contains eighteen different sub-classes for count and mass noun–sense pairs. The distribution of count and mass noun–sense pairs, however, is highly remarkable.

Most linguistic theories of the count–mass distinction follow the assumption – clearly stated in Chierchia (1998a: 56) and subsequent work – that the count–mass distinction is independent of matter. But this assumption raises the issue as to what the linguistic impact of the count–mass distinction actually is. Perhaps one can assume that count expressions are conceptualized as having boundaries, or of coming in "basic" units, and mass expressions as having no boundaries, and not coming in units. The question then is why we find many more nouns that can be classified as count than those being classified as mass. This observation is particularly striking if we further consider that *packaging/sorting* is a productive process, and hence that starting with mass nouns and using packaging/sorting will yield the required count conceptualizations without much ado. There is also a trace of packaging in Chierchia's (this volume, Chapter 2) Type III languages (Yudja, Nez Perce, Indonesian). These languages allow free combination of count *and* mass nouns with numerals but yield container readings when realized in count syntax.[11] In fact, we will show in Section 4 that T4 ambiguity shows the very same effect in English.

There is, however, a further interpretation, to which we will return in Section 5, viz. that count expressions exhibit more structure than mass expressions. To this end, we will assume that at least one way to arrive at a count expression is that it is derived from a mass expression undergoing a process of unit building. This process leads to the imposition of a unit structure on the denotation of the expression, which means in general that structure is added. The present proposal is thus generally in line with approaches like Chierchia (1998a, 2010; this volume, Chapter 2) and Rothstein (2010; this volume, Chapter 3) in assuming that the count–mass distinction reflects a difference in denotational structure of the model to the extent that count senses show more denotational structure than mass senses. In many areas of linguistics, we move from less structure to more structure, but almost never in the other direction. (For example, in discourse, expressions become more specified by context, not less). The very same reason accounts for the pervasiveness of

[11] One could argue that this is a matter of perspective, and that one could similarly argue that Type III languages make no distinction between count and mass. If one takes countability as a feature that distinguishes count from mass nouns, then Type II and III languages will have countability distinctions, but their manifestations are not as in English. However, if one assumes that grammatical features that divide "count" and "mass" nouns for English (such as the plural, numerals, or indefinite articles) are universal, then Type II and Type III languages do not show a count–mass distinction because the grammatical features (which is being assumed to be universal) are not restricted in these languages, as they are in English.

sorting and portioning, and the somewhat obscure status of grinding. The former add structure, the latter, if conceived properly, removes structure. Lexical ambiguity can be used as a probe into the aforementioned process of structure building, which is why it is necessary to consider lexical ambiguity in BECL more closely.

3. Four types of ambiguity in BECL

The core of BECL consists of noun–sense pairs, which by themselves do not make the relation between the count–mass distinction and possible ambiguities apparent. We have thus provided an addition to BECL (to be included in future releases) where all nouns are combined with all annotated senses, and their corresponding countability classes. On the basis of the reorganization of the noun–sense pairs based on noun forms and corresponding senses as well as countability classes, four different types of lexical ambiguity can be identified, as provided in (7).

(7) Types of lexical ambiguity in BECL:
T1: Homonymy
T2: Polysemy without countability shifts
T3: Polysemy with countability shifts
T4: Ambiguity imposed on a single sense by a countability shift

Of the four types listed in (7), the last one, T4 is of particular interest and bears some implications for the analysis of the count–mass distinction as a whole, which we will elaborate on in Section 4. Although the first two types are of no relevance for the count–mass distinction in itself, they are nevertheless relevant insofar as ignoring these types yields awkward results when the pertinent ambiguities – which we assume are T3 and T4 – are discussed (as was observed in our brief discussion of Payne and Huddleston (2002)). Hence, we will briefly discuss all types.

The first type of lexical ambiguity is homonymy (called T1 here), where two or more senses do not bear a discernible semantic relation. The different senses accidentally share one form, and there can be an assignment to different countability classes, as in (8), but this is not mandatory, as illustrated in (9), and in fact is orthogonal to countability.

(8) a. capital#4 (one of the large alphabetic characters used as the first letter in writing or printing proper names and sometimes for emphasis) COUNT
b. capital#2 (wealth in the form of money or property owned by a person or business and human resources of economic value) MASS

(9) a. pen#1 (a writing implement with a point from which ink flows) COUNT
b. pen#2 (an enclosure for confining livestock) COUNT

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Homonymy will not shed direct light on the count–mass distinction because it is a defining feature of homonymy that we find a coupling of unrelated meanings to the same form, as can be witnessed in (8) and (9). The detection of homonymy, however, is important from the perspective of methodology, since homonymy forms pitfalls for theoretical and empirical approaches, as has already been mentioned in the introduction, and again can be stressed with (6). A further extension of BECL 2.1 is a homonymy index, which is particularly appropriate if nouns occurring frequently as count nouns also appear frequently as mass nouns. In fact, no generalization concerning the count–mass distinction can or should be drawn from frequent occurrences of the *nouns* in (6), for the simple reason that we could always be dealing with two different *nouns*. Without further time-consuming scrutiny, the differences cannot be detected. BECL thus provides checks and balances against unwarranted conclusions concerning the count–mass distinction drawn from empirical work where a specific semantic annotation of the nouns in question is not available.

T2 and T3 differ from T1 in that a relationship between the senses can be established, a relationship that can be expressed at different levels of generality, and also at different levels of productivity. The example in (10) provides a T2 ambiguity, i.e. polysemy without a countability shift.

(10)
a. hypothesis#1 (a proposal intended to explain certain facts or observations) COUNT
b. hypothesis#2 (a tentative insight into the natural world; a concept that is not yet verified but that if true would explain certain facts or phenomena) COUNT
c. hypothesis#3 (a message expressing an opinion based on incomplete evidence) COUNT

Of course, T2 ambiguity is of little interest if the count–mass distinction is at stake, since all senses of one noun with T2 ambiguity belong to the same countability class – as is witnessed in (8). Contrary to that is the case of T3 ambiguity, where polysemy also leads to an assignment to different countability classes, as is illustrated in (11).

(11)
a. chicken#1 (the flesh of a chicken used for food) MASS
b. chicken#2 (a domestic fowl bred for flesh or eggs; believed to have been developed from the red jungle fowl) COUNT

When cases like (11) are considered, they are quite often either presented as *dual-life* nouns or as cases of grinding. But the polysemy observed in (9) is not the result of an application of the *Universal Grinder*, which is characterized in (Pelletier 1975) as a thought machine that is aimed at functioning on the cognitive level. The input of the grinder is, however, not a noun, but the denotation of that noun, which is then being ground. The result should be ground stuff of the input. Hence, the grinder turns an entity that is an object to stuff, and presuming that stuff is generally denoted by mass nouns, the output, the ground stuff, should be expressed with a mass term. This does not apply to (11) for the simple reason that beyond the distinction between count and mass, we find a further semantic distinction in that (11a) denotes meat, while (11b) denotes an animal (or a kind of an animal). It is somewhat surprising that the Universal Grinder is actually made responsible for the polysemy in (11), as e.g. in Chierchia (2010). It should be clear that the two senses in (11) are distinct not only in terms of countability but also in their basic characteristics: chickens are not bred for themselves, but for the flesh they provide, and meat does not provide eggs, of course. If the count noun *chicken* would undergo an operation of the Universal Grinder then the outcome would not be chicken meat – which is the actual mass meaning of *chicken* – but ground chicken stuff, which includes chicken bones, plumes, eyes and everything else. This is particularly evident if a formal definition of grinding is provided, as in Rothstein (2010: 392) and discussed above. There is no way to apply Rothstein's definition to arrive at (11a) from (11b). Hence the denotation of the ground term differs both from its input – the countable *chicken#2* in (11b) – and from the denotation of the mass expression in (11a). Grinding in this sense would form an additional sense of *chicken* that (almost) does not exist but is often invoked for the purpose of a funny description of a roadkill.

Recently, Landman (2011) has argued in the same line, suggesting that grinding is not an operation yielding a difference in count and mass, and nothing else, but instead that grinding in most cases is used as a cover term for an unclear semantic relationship between two senses that happen to belong to two different countability classes. The same considerations apply to the Universal Packager (Bach 1986) or the Universal Sorter (Bunt 1985). We argue that countability shifts are much more complex than what can be expressed with grinding or sorting.

As long as the term *dual-life noun* is not properly defined, it is always dangerous to classify a noun like *chicken* with the presented polysemy (which, we have seen, carries over to other animal/meat pairs depending on what is considered edible) as a dual-life noun. The term is also problematic since it seems to describe a lexical property, but it should be clear that the relevant property is actually one of semantic features (cf. (10) above). In Section 4, we will use the term *proper dual-life noun* to delineate nouns showing T4 ambiguity (which is one of countability, and not of other semantic features) from cases of polysemy as discussed here.[12]

Falkum (2010), Husić (2020), and Zamparelli (2020) have proposed generalizations regarding count-to-mass shifts (or vice versa) motivated by the

[12] We make explicit use of the overly specific term *proper* dual-life noun to make clear that many nouns that are usually called *dual-life nouns* are not addressed by this term

regularity of certain cases of polysemy, so-called regular polysemy in the sense of Apresjan (1974).

(12) Regular polysemy (Apresjan 1974, p.16)
Polysemy of the word α with the meanings a_i and a_j is called regular if, in the given language, there exists at least one other word β with the meanings b_i and b_j, which are semantically distinguished from each other in exactly the same way as a_i and a_j, and if a_i and b_i, a_j and b_j are nonsynonymous.

Falkum (2010) thus directly employs such rules of regular polysemy to account for cases like (11). We consider the rules that she proposes to be more appropriate then invoking grinding to cover these cases. The pertinent rules are provided in (13):

(13) a. If an expression has an "animal" use, it also has a "meat"/"fur"/"animal stuff" use.
b. If an expression has a "tree" use, it also has a "wood" use.
c. If an expression has a "fruit" use, it also has a "fruit stuff"/"tree" use. (Falkum 2010: 17)

It should also be noted that such transformations are not restricted to concrete objects; abstract nouns are affected by polysemy, which often results in a shift of countability, too. Husić (2020) argues that abstract nouns are even more sensitive to polysemy and countability shifts and argues that some count senses can regularly be derived from mass expressions which refer to qualities, processes, or states:

(14) if a noun X has a mass sense α which denotes a quality, a process or a state, then it will have a count sense β with one of the possible interpretations:

1. bounded process
2. instances thereof
3. (itemized) placeholders (Husić 2020: 366)

Examples for the three shifts from BECL are provided in (15), (16), and (17).

(15) bounded processes
a. respiration#3 (the bodily process of inhalation and exhalation) MASS
b. respiration#2 (a single complete act of breathing in and out) COUNT

(16) instances thereof
a. inquiry#1 (a search for knowledge) MASS
b. inquiry#2 (an instance of questioning) COUNT

(17) (itemized) placeholders
a. necessity#1 (the condition of being essential or indispensable) MASS
b. necessity#2 (anything indispensable) COUNT

The countability shifts discussed in Falkum (2010) and Husić (2020) cannot be explained in terms of either *grinding* or *packaging*. At this point, we want to

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emphasize the diversity of countability shifts that have tacitly been assumed as the result of universal thought machines. In addition, formal theories of the semantics of count and mass nouns typically focus on the mere shift between count and mass and tend to ignore the changes in interpretation that occur with them (cf. Chierchia 2010: 128–30; Rothstein 2010: 390), which again proves that a fine-grained distinction of different types of ambiguity is necessary for a proper assessment of the count–mass distinction. Let us now turn to T4 ambiguities.

4. T4 Ambiguities: (Almost the) Same Meaning, Different Countability

T4 ambiguity is characterized by a single sense of a noun that allows *both* count and mass interpretations, where the count interpretation differs only in a single aspect from the mass interpretation, which can be characterized by individualizing the mass interpretation. (We tacitly assume that the mass interpretation is logically prior to the count interpretation, an assumption that will be discussed in more detail below.)

Since this will play a role in Section 5, we would like to explicitly point out the reactions of the annotators towards nouns showing T4 ambiguities. Annotators were asked to embed the pertinent noun in four syntactic contexts, and to further provide semantic estimates if certain of the embeddings were possible. The four syntactic contexts are as follows:

– Syntax 1: [more N_{sg}] Can the noun appear with *more* when realized in singular number?
– Syntax 2: [more N_{pl}] Can the noun appear with *more* when realized in plural number?
– Syntax 3: [[$_{Det}$ indef] N_{sg}] Can the noun appear with an indefinite determiner?[13]
– Syntax 4: [ø N_{sg}] Can the singular noun appear without a determiner?

We would expect that mass nouns may appear in Syntax 1, while count nouns may appear in Syntax 2. Similarly, count nouns should appear in Syntax 3, while mass nouns should appear in Syntax 4. The first two and the second two syntactic contexts can thus be viewed as quasi-opposites. An affirmative answer to either Syntax 1 or Syntax 2, however, may be due to a meaning shift, which we must detect, in order to not conflate different senses. With regard to Syntax 1, an affirmative answer may have been provided with a comparison in mind that does not address individual items. With regard to

[13] Since English does not show indefinite plural determiners, the context is tantamount to assuming that the noun appears in singular number.

Syntax 2, an affirmative answer could have come about because the annotator had a hidden container reading in mind. Thus, affirmative answers to Syntax 1 and Syntax 2 will lead to further asking Semantics 1 or Semantics 2, respectively:

- Semantics 1: If Syntax 1 is answered affirmatively: Is the comparison based on counting individual items or on some other mode of measurement?
- Semantics 2: If Syntax 2 is answered affirmatively: Is the plural equivalent to a hidden classifier reading (*kinds of*, *containers of*, *types of*)?

T4 ambiguities are remarkable in that all four syntactic contexts are answered affirmatively, and the two semantic clarifications are answered negatively, i.e. the noun appears in Syntax 1, but the comparison is not based on individual items (as opposed to typical fake mass nouns such as *furniture* or *lingerie*; cf. Kiss, Pelletier, and Stadtfeld (2014)); and the noun also appears in Syntax 2 but there is no hidden classifier reading. The general position in discussions of the count–mass distinction is to expect opposite reactions to Syntax 1 and 2, and if the reactions are not opposite, we expect that either Semantics 1 or 2 is answered affirmatively. In these cases, the nouns are classified as either mass or count nouns. But in the present case (and also with affirmative answers to Syntax 3 and 4), the nouns are classified as being *both* count and mass nouns.

The definitions of some of the nouns falling in this class given in WordNet are provided in (18).

(18)
a. brunch#1: combination breakfast and lunch; usually served in late morning
b. candy#1: a rich sweet made of flavored sugar and often combined with fruit or nuts
c. rainfall#1: water falling in drops from vapor condensed in the atmosphere
d. cake#3: baked goods made from or based on a mixture of flour, sugar, eggs, and fat
e. casserole#1: food cooked and served in a casserole
f. salad#1: food mixtures either arranged on a plate or tossed and served with a moist dressing; usually consisting of or including greens
g. swamp#1: low land that is seasonally flooded; has more woody plants than a marsh and better drainage than a bog
h. confirmation#1: additional proof that something that was believed (some fact or hypothesis or theory) is correct
i. contradiction#3: the speech act of contradicting someone

In order to reproduce the count and mass uses of the nouns in (18) we provide examples from the Contemporary Corpus of American English (COCA; Davies 2009), in which we present the mass interpretation in the a)-sentences, singular count interpretations in the b)-sentences, and plural count interpretations in the c)-sentences.

(19) a. He would hire me for brunch shifts because most cooks hated doing *brunch*.
b. We sleep till noon, have the maid cook *a brunch* and a snack before each meal.
c. But only on Sundays does it serve *its superb brunches*.

(20) a. The EU response includes *deployment of naval vessels and surveillance planes* to the area.
b. The captain was midway through *a deployment in the Mediterranean*.
c. In the last decade, there have been *twenty-seven army deployments*.

(21) a. In 1935, it became clear why *emigration* was halted.
b. During 2007, there was *a massive emigration* of Arabs from the Palestine territories.[14]
c. I survived *two emigrations*.

(22) a. There was *major looting*, no ifs, ands, or buts about it.
b. The fracas turned into *a mob looting* that lasted till 1 a.m.
c. In all, county police are investigating *40 lootings*.

(23) a. The international development community wanted Amazonians to cut down *rainforest*, plant pasture, and raise cattle.
b. REDD could protect one rainforest from clear-cutting merely by sending the logging trucks to *a different rainforest*.
c. The Asian tree frog lives in *the moist rainforests* of India.

(24) a. There is *swamp* where Florida should be.
b. We emerged from *one swamp* and landed in another.
c. The river was obscured by *two great swamps*.

It should be clear now that cases of T4 are pure (in fact: the only) instances of what is called *dual-life nouns*. As T3 cases come together with a shift and/or addition of meaning, there is only a single meaning change affecting T4 cases, which leads to an individuation of the mass meaning, as can be initially characterized by the definition in (25):

(25) T4 ambiguity: If *X* is the mass interpretation, when used as a count noun, the interpretation becomes *an individual X*.

Given the different semantic types of nouns involving T4 ambiguity, it becomes necessary, however, to further specify what is meant and allowed by (25). The first case to be considered can be illustrated with *brunch*. This noun is already ambiguous, in that it can mean the aggregate food that can be eaten, or the event during which the food is eaten.[15] We will call this a stuff/event ambiguity. Note that the ambiguity is not affected when *brunch* undergoes T4 realizations, i.e. the event and the stuff sense show up on both sides of the count–mass distinction, and the count sense can be characterized as an

[14] This can be a closed aggregate of individual instances but can also be an individual instance.
[15] WordNet does not draw this distinction, but it becomes obvious once a sufficient number of examples is considered.

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individuation. In (19), we see the stuff sense; hence (26) provides event senses showing the same T4 ambiguity:

(26) a. *Brunch* can take a while so start slowly with a fresh fruit platter.
b. On Sunday, the Renaissance Waverly Hotel hosts *a brunch* with Mr. and Mrs. Claus.
c. Weekend dances and *Sunday brunches* became fixtures in the postwar era.

The second aspect to note is that *individuation* does not necessarily mean that we are talking about single individuals. This has already been illustrated with *brunch*, which – in the pertinent sense – can be characterized as an aggregate of food, which can be individualized *as an aggregate* (the same point can be illustrated with *rainforest* and *emigration*). This holds for event interpretations as well, with the addition that the nouns allow genuine ambiguity with regard to an aggregation of individual events, an ambiguity that is not necessarily resolved, as can be witnessed by (21b), where *emigration* may refer to an aggregate of emigrations, or to a single event.

Of course, these examples also show that nouns expressing T4 ambiguity do not have to refer to concrete entities, but allow *event-like* structures, and may also denote abstract entities, as is witnessed by *friendship* in (27).

(27) a. He has been tapped to compose a speech promoting *friendship* between Washington and the "Muslim world."
b. He did have *a close friendship* with the warden.
c. My brother had *very long friendships*.

A possible further property of these nouns will be noted with some caution: the corpus data revealed that the count instances more often than not are realized in the plural, but that numerals occur rarely (as, e.g., illustrated in the c)-examples of (20), (21), (22), and (24)). After initially realizing that the extracted samples showed numerals only rarely, we explicitly searched for numerals but found no further occurrences with these nouns. The average proportion of numerals occurring with these nouns is extremely low, a mere three percent. We will return to this issue in Section 5.

To sum up, nouns showing a T4 ambiguity are both count and mass, with the only difference in meaning stemming from an individuation. The nouns in question can have concrete and abstract interpretations. Individuations may yield single individuals, but also single aggregates. In the following section, we will discuss the implications of T4 ambiguities for theories of the count–mass distinction.

5. Implications for Count–Mass Theories

BECL shows an interesting distribution of count and mass noun senses: the majority of noun sense types show unique count syntax, i.e. annotators have uniquely classified the pertinent noun–sense pairs to occur in syntactic count contexts. A smaller group uniquely occurs in mass syntax. If we compare these findings with the major strands of current research on the count–mass distinction, they seem to suggest that *not all nouns are count and mass at the same time*. This contradicts the account found in Borer (2005) and Pelletier (2012), where semantically neutral nouns are realized in given syntactic contexts and become count or mass at a phrasal level.

T4 ambiguities have further shown that the class of nouns that are count and mass at the same time is rather small. These nouns can appear in both count and mass syntax, and apart from the introduction of an individuation, the senses used in count and mass syntax cannot be discerned. We assume that nouns showing T4 ambiguities are the only nouns that can be classified as proper *dual life nouns*.[16]

Pelletier (1975; 2012) has argued that grinding can be used to propose that nouns are count and mass at the same time. The findings of BECL do not support this claim: To begin with, it remains rather unclear which phenomena should be encompassed by grinding. Following Falkum (2010) and Landman (2011), we have pointed out that putative cases of grinding show differences in meaning and can be captured by rules of polysemy as well – thus putting these cases in the class of T3 ambiguities, where a discernible but related semantic difference is associated with a given noun form if it is realized either in count or in mass syntax. T4 nouns, on the other hand, show no trace of *grinding* effects. With regard to *grinding* in general, we note that mass-to-count conversions (Universal Sorter, Universal Packager) appear to be much more productive than count-to-mass conversions (Universal Grinder), most illustrations of which are either freely constructed, or drawn from puns or advertisements. Katz and Zamparelli (2012: 374) seem to contradict our claim. They state that the frequent mass use of "'obvious' count nouns such as *car*, *ski*, and *piano* seem surprising."

(28) a. How much car can you afford?
b. Phantom 87s were too much ski for me.
c. I always practice too much piano.

Yet, they also conclude that "the meaning shifts which seem to take place when a count form is used as a mass are a lot more diverse than [the simple version of universal grinding of the material]."[17] They also admit that other syntactic and semantic factors may play a role, that e.g. *practice too much*

[16] We note that this BECL class contains at least one celebrity noun, which is typically used to illustrate dual life nouns, viz. *cake*.
[17] Of course, we would not speak of count *forms*, since we assume that the count–mass distinction applies at the level of the sense (but see footnote 18 below).

piano in (28c) could be a reduction from *practice too much piano playing*, where mass syntax does not appear to be surprising since *playing* is the actual (reduced) head of the construction, and of course *mass*. Cases like (28a) are found quite often in advertisements, particularly advertisements referring to possible affective reactions, which also plays a role in (28b). All this shows that the interpretations of the nouns are affected in more than one way, basically yielding different interpretations, which are independent of the count–mass distinction, but of course may interact with it.

With regard to T4, we note that the parallel appearance of *more* with a singular and plural occurrence of pertinent nouns does not lead to meaning shifts. For a noun like *swamp*, the mode of measurement is not based on number when it shows up as part of the phrase [$_{NP}$ more swamp], and it does not induce a hidden type or packaging reading when used as part of the phrase [$_{NP}$ more swamps].

Given these observations, an analysis of the count–mass distinction that relies on a semantic difference between count and mass nouns seems to us more plausible. Such a proposal would clearly account for the clear-cut classifications for the majority of nouns in BECL (which holds for 93% of the annotated noun types, with count nouns amounting to 72%, and mass nouns to 21%).

A large group of the remaining 7% consists of noun senses that are classified as being neither mass nor count, because they generate negative answers to all four syntactic context questions. Some thirty-eight of these senses were proper nouns or what we have called "quasi-proper nouns" (where other senses of these lemmas fall into other, more ordinary groups of senses), as illustrated in (29).

(29) a. milk#3 (a river that rises in the Rockies in northwestern Montana and flows eastward to become a tributary of the Missouri River)
b. devil#1 ((Judeo-Christian and Islamic religions) chief spirit of evil and adversary of God; tempter of mankind; master of Hell)

Leaving these cases aside, we are left with a number of rather common noun senses that do not show what we have taken to be the characteristic properties of count or mass meanings. A few examples are provided in (30).

(30) a. midair#1: some point in the air; above ground level; "the planes collided in midair"
b. nature#3: the natural physical world including plants and animals and landscapes, etc.; "they tried to preserve nature as they found it"
c. unison#1: corresponding exactly; "marching in unison"
d. open#2: where the air is unconfined; "he wanted to get outdoors a little"
e. open#4: information that has become public; "all the reports were out in the open"

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f. limelight#1: a focus of public attention; "he enjoyed being in the limelight"
g. dollar#4: a symbol of commercialism or greed; "he worships the almighty dollar"
h. midst#1: the location of something surrounded by other things; "in the midst of the crowd"
i. heyday#1: the period of greatest prosperity or productivity
j. bitch#1: an unpleasant difficulty; "this problem is a real bitch"

It is not so clear to us how these noun senses will fit into a fully unified account of the count–mass distinction. We hence have to leave this issue to future research.

Coming back to the bulk of the count and mass senses, a proposal should also answer why there are many more count than mass noun types (count noun types outnumber mass noun types by a factor of 3.5 in BECL), in addition to providing a place for T4 ambiguities. Both Chierchia (1998a; 2010; this volume, Chapter 2) and Rothstein (2010; this volume, Chapter 3) have argued that there is a denotational distinction between count and mass nouns that is reflected in their syntax. Rothstein (2010; this volume, Chapter 3) explicitly assumes that count and mass nouns are of different types ($<e \times k, t>$ and $<e, t>$, respectively, where k provides a context in which individuals count as atoms). Rothstein further assumes that count and mass denotations are not independent of each other but instead that count denotations are derived from mass denotations by providing a context in which semantic atoms are present. Similarly, Chierchia (1998a, 2010; this volume, Chapter 2) shows that various denotational constraints (as, e.g., whether something is a property or a kind) interact to distinguish count and mass nouns, which at the same time have common roots.

Chierchia (this volume, Chapter 2) suggests a distinction in three types of languages when it comes to the count–mass distinction:

– Type I languages allow count nouns to appear in count syntax.
– Type II languages require additional means to allow nouns to occur in count syntax.
– Type III languages allow all nouns in count and in mass syntax.

Let us combine our findings with Chierchia's typology. With regard to English, we note that the majority of noun senses have been classified as count senses, and hence the nouns bearing the senses are classified as count nouns.[18]

[18] In fact, it is even correct to claim that BECL mostly contains count *nouns*. For the currently unreleased version 3.0, we have transformed the entries into a lexicon of nouns, where each noun is assigned *a set of meanings and the corresponding set of countability classes*. This information is used to compile classes across senses and countability assignments. The most frequent classes according to this classification are unique count nouns (4,493 entries) and unique mass nouns (1,257 entries). It is remarkable that the ratio between count and mass nouns equals the ratio between count and mass senses.

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A minority, however, behave as if it belonged to a Type III language, and hence allow appearances in count and mass syntax equally well. Here we assume that Chierchia's meaning shift from mass denotations to containers is equivalent to what we call individuation.

Now, there are many more count noun types than mass noun types in BECL. So, there are many nouns that either exclusively occur in count syntax or in mass syntax. What is more, we note that mass-to-count shifts occur more often than count-to-mass shifts. And finally, we observe T4 ambiguities with a small class of nouns in English. If one takes Rothstein's (2010; this volume, Chapter 3) denotational system as a starting point, one notes that mass expressions are simpler than count expressions typewise. It would, however, be a somewhat bold proposal to conclude from this denotational difference that mass expressions are logically prior to count expressions.[19] Instead, let us assume that expressions are more or less likely to start their existence as showing up in count or mass contexts. With regard to count and mass shifts, Rothstein (2010: 391) considers count-to-mass shifts more problematic than mass-to-count shifts. Her analysis, however, invokes a reference to *natural atomic predicates* in addition to being *semantically atomic predicates*. Following Chierchia (1998a), she further assumes "that type shifting will be blocked unless the resulting interpretation is sufficiently different from interpretations that are otherwise available" (Rothstein 2010: 391). In light of these suggestions, we may conclude that T4 ambiguity ensues because an individuation leads to count syntax, and to an enriched denotational structure (in other words: individuals become available, once again, possibly through providing semantic atoms, but this could also be achieved by employing the system provided by Chierchia (this volume, Chapter 2)). Consequently, the nouns can actually be counted.

In any case, we assume that structure is added to the interpretation of a noun. This is the basic difference that we can observe in T4 ambiguities: nouns like *swamp* start out as mass expressions but then become individualized units (which in itself, as the discussion of *brunch* has shown, do not have to be atomic individuals). The additional structure may eventually lead to a petrification of the count sense, which in terms of Rothstein's (2010) analysis might amount to the assumption that semantically atomic predicates are taken to become natural atomic predicates. For this reason, there are many more count than mass nouns: there is nothing that forces a noun to add denotational structure, but once this has happened, there is no way back.

[19] We take it, for instance, as highly unlikely that *tables* was ever used as a mass expression. Although we obviously do not endorse an ontological analysis of the count–mass distinction, we cannot but acknowledge that certain aspects of the count–mass distinction remain unaccounted for in all approaches unless one either accepts hidden circular explanations or an ontological base.

This conclusion will also shed light on the distribution of grinding and packaging (or sorting). As we have pointed out already (just as Rothstein (2010) did), grinding is less productive than packaging and sorting and also comes with strings attached insofar as they show pun-like qualities and other forms of metaphor and metonymy. Grinding, of course, was always considered to *remove* structure, to turn some individual into a bunch of goo (to quote George Patton here). What we have argued is that such a removal of structure goes against the grain of language. Expressions can be (comparatively) unstructured – as is shown by proper mass nouns – but once they have shown a count sense (either as an additional sense without further changes in meaning, such as T4 ambiguities, or after petrification and addition of meaning, so that the mass sense is gradually lost), their denotational domain provides more structure, and moving from more to less information is generally not favored by language.

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